# Hyunsoo Lee

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## About me

My name is Hyunsoo Lee, and I am currently pursuing a Ph.D. in Professor Jeyavijayan (JV) Rajendran's lab at Texas A&M University, focusing on hardware vulnerability detection. During my master's program, I worked in the VLSI CAD and design domain, researching algorithms to maximize and automate the performance of analog and digital circuits. Now, building on broad VLSI expertise, I am exploring optimal methodologies for detecting vulnerabilities that can arise in real processors and beyond the hardware level.

I am currently seeking a hardware verification internship in the U.S. I am in F-1 student status and can be authorized to work through Curricular Practical Training (CPT) for an internship, so employer visa sponsorship is not required.

## RESEARCH INTERESTS

**Hardware Security & Vulnerability Detection**
- Hardware vulnerability detection in System-on-Chip
- PCIe bug detection using hardware fuzzing

**VLSI Computer-aided Design (CAD)**
- CAD algorithms for optimizing VLSI design
- Full design automation of analog & digital circuits

**VLSI Hardware Design**
- Analog & Digital mixed system design
- Compute-in-Memory (CIM) and Compute-Near-Memory (CNM)

## SCHOLARSHIPS / AWARDS

**Scholarship for Academic Excellence** — 2017, 2020-2022, 2024
*Kyungpook National University, Korea*

**International SoC Design Conference (ISOCC) – 2023 AI Semiconductor Forum Award** — Oct. 27, 2023
*Awarded as the best paper for "High-throughput PIM (Processing in-Memory) for DRAM using Bank-level Pipelined Architecture"*

## EDUCATION

**Texas A&M University** — Sep. 2025 - Present
Ph.D. in Electrical and Computer Engineering
Advisor: Jeyavijayan (JV) Rajendran
Secure and Trustworthy Hardware (SETH) Lab.
https://seth.engr.tamu.edu/

**Kyungpook National University (KNU),** *Daegu, South Korea* — Sep. 2023 - Aug. 2025
M.S. in Electronic and Electrical Engineering
Advisor: Taigon Song
Intelligent 3-Dimensional Very Large-Scale Integrated Circuits (I3D VLSI) Lab.
https://i3dvlsi.wordpress.com/
Cumulative GPA: 3.88/4.3

**Kyungpook National University (KNU),** *Daegu, South Korea* — Mar. 2017 - Aug. 2023
B.S. in Electronics Engineering
Cumulative GPA: 3.55/4.3 (Major: 3.79/4.3)

## PROFESSIONAL EXPERIENCE

**Ph.D. at SETH Laboratory (Prof. Jeyavijayan "JV" Rajendran)** Sep. 2025 - Present
*Dept. of Electrical and Computer Engineering at Texas A&M University*

- **Autonomous Hardware Vulnerability Detection in PCI Express**
  - Constructing PCIe fuzzing framework to detect security vulnerability in autonomous manner (in progress)

- **Assessing Effective Hardware Verification Techniques in System-on-Chip (SoC)**
  - Derive the most efficient methodology for SoC security vulnerability detection by comparing approaches using OpenTitan, a modern SoC platform (in progress)
  - Include manual detection, formal verification, random regression testing, fuzzing, information flow tracking (IFT), static linting, and LLM-based approaches for the methodologies

**M.S. at I3D VLSI Laboratory (Prof. Taigon Song)** Sep. 2023 – Aug. 2025
*Dept. of Electronic Engineering at Kyungpook National University (KNU), Daegu, South Korea*

- **Design Automation/Optimization of LC Voltage-Controlled Oscillator (LC-VCO) circuit**
  - Established a Complete LC-VCO auto-design framework with physical layout using Reinforcement Learning and Gradient Descent Algorithm
  - Developing process-portable LC-VCO layout auto-generator

- **Tape-out AES-128 with Hardware Trojan Application-Specific Integrated Circuit (ASIC)**
  - Designed 4 different chips with and without Hardware Trojan in one die in 1st tape-out
  - Designed 2 different chips of analog and digital mixed system in 2nd tape-out

- **Back-side Power Delivery Network (BSPDN) Optimization**
  - Established BSPDN design flow and analysis methodology using commercial EDA tools
  - Developed BSPDN optimization framework using Reinforcement Learning

- **Low-power AI Accelerator based on Analog Computing in Memory using Resistive Random Access Memory (RRAM)**
  - Managed project team members aiming to achieve low-power specification
  - Simulated whole design with analog and digital mixed system

**Undergraduate Intern at I3D VLSI Laboratory (Prof. Taigon Song)** Sep. 2021 - Aug.2023
*Dept. of Electronic Engineering at Kyungpook National University (KNU), Daegu, South Korea*

- **Processing-in Memory Using DDR5 SDRAM**
  - Designed Processing-in Memory architecture using Synopsys EDA tools
  - Completed Place & Route process of top-level layout using DRAM macro cells
  - Automated PIM layout design using shell scripts

## PUBLICATIONS

- Yunjeong Shin, Daehyeok Park, Dohun Koh, Dongryul Heo, Jieun Park, Hyundong Lee, Jongbeom Kim, **Hyunsoo Lee**, and Taigon Song, "FS2K: A Forksheet FET Technology Library and a Study of VLSI Prediction for 2nm and Beyond," IEEE International Symposium on Circuits and Systems (ISCAS), 2024

- Seungmin Woo, **Hyunsoo Lee**, Yunjeong Shin, MinSeok Han, Yunjeong Go, Jongbeom Kim, Hyundong Lee, Hyunwoo Kim and Taigon Song, "Reinforcement Learning-Based Optimization of Back-side Power Delivery Networks in VLSI Design for IR-drop Reduction," Design, Automation and Test in Europe Conference (DATE), 2024
- **Hyunsoo Lee**, Minseung Shin, Gyuri Shin, Sumin Jeon, Hyundong Lee and Taigon Song, "High-throughput PIM (Processing in-Memory) for DRAM using Bank-level Pipelined Architecture," IEEE International SoC Design Conference (ISOCC), 2023 (AI Semiconductor Forum Award)

## SKILLS & CERTIFICATIONS

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|---|---|
| **Verification tools** | Cadence Jasper Gold, Synopsys VC Formal |
| **EDA tools for VLSI design** | Cadence {Virtuoso/EMX/Innovus}, Synopsys {Custom Compiler/Design Compiler/IC Compiler II/Fusion Compiler/StarRC/ PrimeTime/Formality/PrimeLib} Ansys {Redhawk} |
| **Simulators** | HSPICE, PrimeSim, Verdi, VCS, Spectre |
| **Developer Tools** | Vim, gVim, VS Code |
| **Programming** | Shell script, TCL, Python, SKILL, LATEX |

## TEACHING EXPERIENCE

**Teaching Assistant for Digital IP Intensive – Place&Route** Nov 2023
*Samsung, Gyeonggi-do, Republic of Korea*

**Teaching Assistant for Digital Integrated Circuit (ELEC0413-002)** Fall 2023
*Kyungpook National University (KNU), Daegu, South Korea*